

An Elo Rating System for NFL Pass Rushers & Blockers

Maximilian Gebauer^{1,2} Victoria Lee^{1,3} Kenny Watts²

¹School of Arts and Sciences, University of Pennsylvania

²The Wharton School, University of Pennsylvania

³School of Engineering and Applied Science, University of Pennsylvania

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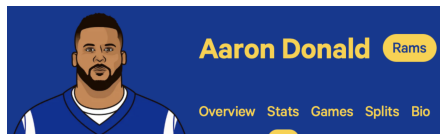
Outline

- Background on Current Player Evaluation
- Issues with Current Approach
- Model Overview
- Ridge Model
- Conclusion

Current evaluation methods

- **Pass Block Win Rate:** Pass rusher performance is conventionally assessed using *pass block win rate*, defined as the proportion of pass rush attempts in which the rusher beats their blocker within 2.5 seconds.
- **Aggregated Statistics:** Simple counts of sacks, hits, or hurries are used to rank rushers/blockers
- **Issues:**
 - No adjustment for matchup difficulty (player quality or double-teams)
 - No differentiation of event (hit, sack, or hurry) impact on game
 - Measures are not dynamic within a season (insensitive to non-stationarity in player skill)

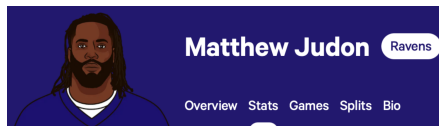
Motivating Example



Aaron Donald Rams

Overview Stats Games Splits Bio

Stats: 17 GP, 38 TKL, 46 AST, 84 COMB, 19 TFL, 12.5 SCK



Matthew Judon Ravens

Overview Stats Games Splits Bio

Stats: 17 GP, 38 TKL, 22 AST, 60 COMB, 14 TFL, 12.5 SCK

Season	TM	GP	TKL	AST	COMB	TFL	SCK
2021	LAR	17	38	46	84	19	12.5

Season	TM	GP	TKL	AST	COMB	TFL	SCK
2021	NE	17	38	22	60	14	12.5

- How can we non-arbitrarily evaluate the quality of these players with such similar aggregate stats?
 - How can we quantify the skill-disparity, should it exist?

Research Questions

- How can we incorporate matchup difficulty at the individual play level to better evaluate pass rusher ability?
- How can we dynamically assess pass rusher performance on a play-by-play basis to identify temporal trends or shifts in skill?
- Which evaluative metrics are most suitable for comparing the effectiveness of dynamic, play-level approaches against traditional static evaluation models?

Introduction to Elo Rating

Elo rating is a widely-used method for assessing player or team skill levels, initially developed for chess.

Example (Chess):

- Players are assigned numerical ratings representing their skill levels.
- Ratings adjust based on game outcomes and opponent strength.

Probability of Player A winning:

$$P(\text{A wins}) = \frac{1}{1 + 10^{(R_B - R_A)/\alpha}}$$

Definitions:

- R_A : Rating of Player A before the game
- R_B : Rating of Player B before the game
- α : Scaling parameter (conventionally 400 in Chess)

Implied Win Probability across Elo Differences

Winning Probability of Player A vs. Opponent Elo

Player A Elo fixed at 1400; Player B Elo varies from 800 to 2000

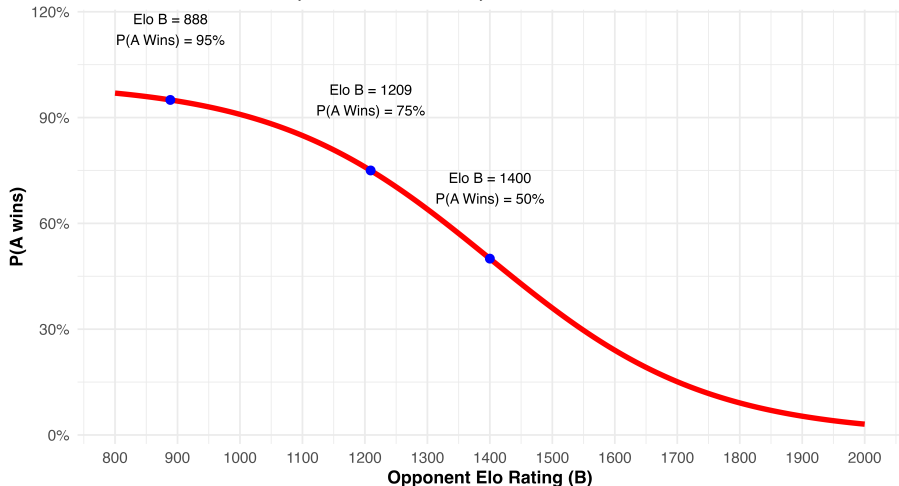


Figure: Scaling of win probability depending on opponent Elo rating

How the Model Works

We sequentially iterate through each interaction in a player's history to update their Elo score based on each matchup outcome

Rating Update:

$$R'_A = R_A + K_t(S_A - E_A)$$

Win Probability:

$$P(\text{A wins}) = \frac{1}{1 + 10^{(R_B - R_A)/\alpha}}$$

Where S_A = observed outcome, E_A = expected outcome, and K_t = learning rate at time t

Data Processing

- Data was acquired from Hudl for the full 2021-2022 season with labeled player matchups and play outcomes
- After calculating Euclidean distance between all linemen and the quarterback, outcomes were labeled as so:
 - **Beat:** $d_{def} < d_{off}$ before t_{pass} , where each distance is the distance between that side's lineman and the quarterback
 - **Hit:** Any event outcome labeled as a tackle occurring behind the line of scrimmage before the pass without a sack label
 - **Sack:** Already labeled in dataset
- We also label any double teams where blocker engagements overlapped for any given rusher

Elo Adjustments

- We implement a multinomial Elo model by weighing each outcome based on relative impact on EPA (?), with the greatest impact given to sacks
- Offensive linemen involved in a double team receive a temporary 100 point bonus for the interaction ($\approx 10\%$ increase in implied WP), resulting in less reward for winning and greater penalty for losing
- Symmetrically, defensive lineman facing a double team face proportionally smaller penalty for losing and greater reward for winning

Adaptive Learning Rate

We implement an **adaptive learning rate** to learn player ability faster in initial interactions, then slowly decrease throughout the season

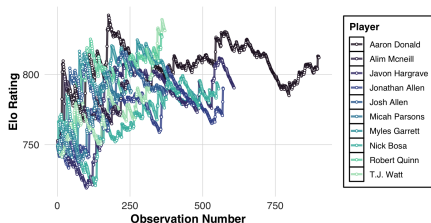
$$K(n) = \begin{cases} K_{\text{start}}, & \text{if } n \leq n_{\text{prov}} \\ K_{\text{start}} + (K_{\text{min}} - K_{\text{start}}) \cdot \frac{n - n_{\text{prov}}}{n_{\text{decay}} - n_{\text{prov}}}, & \text{if } n_{\text{prov}} < n \leq n_{\text{decay}} \\ K_{\text{min}}, & \text{if } n > n_{\text{decay}} \end{cases}$$

where we set n_{prov} at 90 and n_{decay} at 210 interactions

Results - Looking At Top Performers

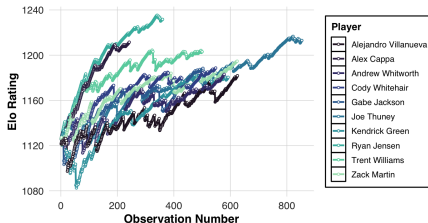
Top 10 Rushers: ELO Trajectories

Evolution of Elo rating over successive plays



Top 10 Blockers: ELO Trajectories

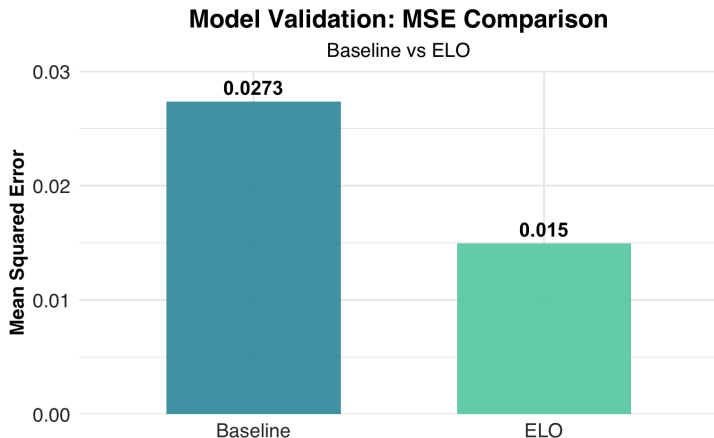
Evolution of Elo rating over successive plays



- 7 Rushers and 3 Blockers Selected to the Pro Bowl in 2021
- In line with pre-theoretic expectations

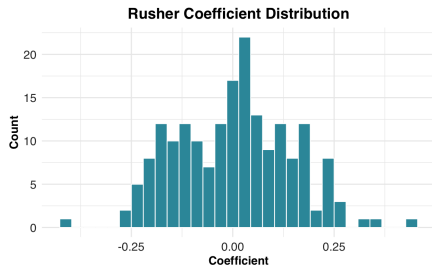
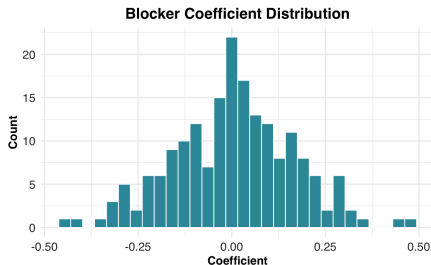
Results - Model Validation

We compare win probability given by Elo score differences with our naive pass block win rate probabilities and see that our model performs better



Ridge Regression

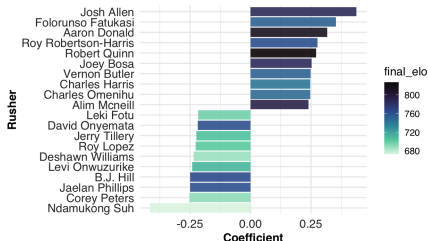
- Since Elo converges to a logistic regression model in the limit, assuming stationarity in skill, we considered a logistic model with $\mathcal{L}_2$ penalty
- We see that the coefficients are distributed about 0, providing an intuitive benchmark for ability



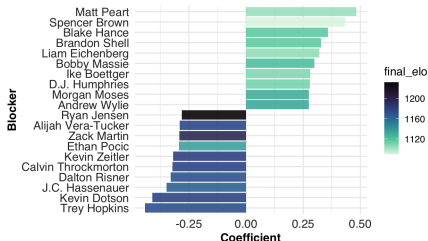
Results - Comparison of Top Players

- We can analyze players with the most negative and positive coefficients
- Since our Ridge model predicts rusher wins, higher coefficients are more desirable for rushers whereas lower coefficients are more desirable for blockers

Top and Bottom 10 Rushers by Ridge Coefficient



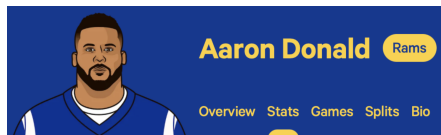
Top and Bottom 10 Blockers by Ridge Coefficient



Results - Discussion

- Elo gives a dynamic view of a player's skill level throughout a season, but can have high variance for within season predictions
- Ridge centers at 0, which we can label as the "average" player but it loses the dynamic aspect of temporal rating systems given it assumes stationarity in skill
- While the two models identify generally different top players, we do see broad alignment in their ratings
- Takeaway: Ridge optimizes for binary prediction in the short-term, Elo prioritizes player-rating and identifies non-stationarity

Returning to Motivating Example



Aaron Donald Rams

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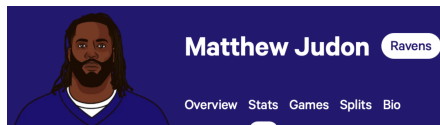
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Season	TM	GP	TKL	AST	COMB	TFL	SCK
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Elo of **872**

Probability of Beating an Average
Blocker: **16.9%**

Ridge Coefficient: **0.317**



Matthew Judon Ravens

Overview Stats Games Splits Bio

Stats: GP 17, TKL 38, AST 22, COMB 60, TFL 14, SCK 12.5

Season	TM	GP	TKL	AST	COMB	TFL	SCK
2021	NE	17	38	22	60	14	12.5

Elo of **793**

Probability of Beating an Average
Blocker: **10.3%**

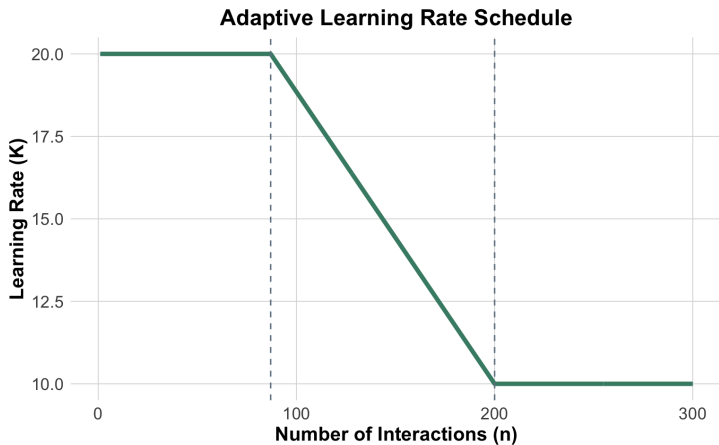
Ridge Coefficient: **0.012**

Conclusion

- Summary of contributions
- Future work
 - ① Compare Elo to adjacent dynamic models (Glicko, Glicko-2)
 - ② Consider extensions to run blocking (RBWR)
 - ③ Detect and account for RB and TE chips
 - ④ Consider varying scaling parameter α with magnitude of ratings to improve prediction accuracy
- Acknowledgments: Our gratitude to Matt Edwards (Hudl) for providing the data and to Dr. Abraham Wyner, Dr. Paul Sabin, and Jonathan Pipping for their invaluable advice, mentorship, and assistance

References I

Appendix - Adaptive Learning Rate

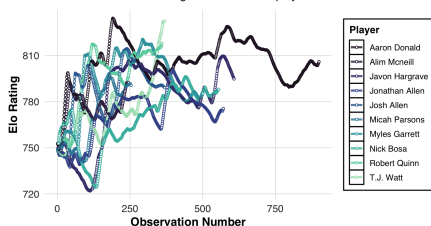


Appendix - Smoothed Trajectories

We see that smoothing over 20 interactions leads to lower sensitivity to play by play interactions and less noisy trajectories

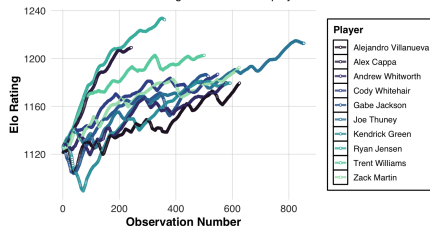
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Top 10 Blockers: ELO Trajectories

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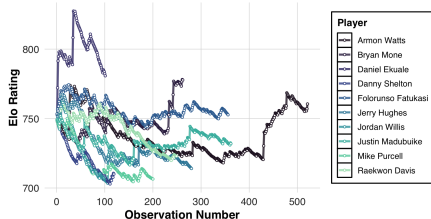


Appendix - Trajectories of Random Sample

We see that the random trajectories show less improvement over time than our top players, showing the ability of Elo ratings to accurately separate players by skill level

Random 10 Rushers: ELO Trajectories

Evolution of Elo rating over successive plays



Random 10 Blockers: ELO Trajectories

Evolution of Elo rating over successive plays

